26900369 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours) REGULAR EXAMINATION October 2026

Fifth SEMESTER

Discipline Specific Core Course (DSC) - MG5DSCMBG300 - AGRICULTURAL MICROBIOLOGY

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Notes Questions

Answer any **10** questions

Each question carries **2** marks

1. Give four examples of soil algae. [U] / [CO1]
2. How does soil pH affect bacterial and fungal growth? [K] / [CO1]
3. Define predation. [K] / [CO1]
4. Define synergism with suitable example. [K] / [CO1]
5. Differentiate between phosphorus mineralization and immobilization. [K] / [CO2]
6. What are epiphytic and endophytic microorganisms? [K] / [CO2]
7. Distinguish between microbial retting and chemical retting. [An] / [CO2]
8. What is meant by root exudate? [K] / [CO2]
9. How is sugarcane mosaic disease transmitted? [U] / [CO3]

10. Why does soft rot spread rapidly under warm and humid conditions? [U] / [CO3]

11. Summarize how applying lime to the soil helps control club root disease. [U] / [CO3]

12. State the main types of biopesticides. [K] / [CO3]

[2x10 = 20]

Part B

Short Essay Questions

Answer any **4** questions

Each question carries **5** marks

13. Explain in detail about *Trichoderma* as a biocontrol agent. [A] / [CO3]

14. Illustrate how Rhizobium inoculation can be used to improve the growth of a leguminous crop. [A] / [CO3]

15. Explain mutualism with suitable examples. [U] / [CO1]

16. Distinguish between ectomycorrhiza and endomycorrhiza. [U] / [CO1]

17. Describe the role played by microorganisms in the process of developing tobacco leaf quality during tobacco curing. [K] / [CO2]

18. Explain the role of respiration in carbon cycle. [A] / [CO2]

[5x4 = 20]

Part C

Essay Question

Answer any **1** questions

Each question carries **10** marks

19. Explain in detail nitrogen cycle with the help of a diagram. [A] / [CO2]

20. Explain how structural and biochemical defence mechanisms work together to protect plants from microbial infection. [A] / [CO3]

[10x1 = 10]